

Detailed Solutions to Quiz 3

Randomized Algorithms (2026 Spring)

Problem 1. Alice and Bob play n independent games, and Alice wins each game with probability 0.6. Use a Chernoff bound to upper-bound the probability that Alice loses the tournament.

Solution. Let

$$X_i = \begin{cases} 1, & \text{if Alice wins game } i, \\ 0, & \text{otherwise.} \end{cases}$$

Then $X_1, \dots, X_n$ are independent Bernoulli random variables with

$$\Pr[X_i = 1] = 0.6.$$

Let

$$X = \sum_{i=1}^n X_i.$$

Then X is the total number of games won by Alice, and

$$\mu := \mathbb{E}[X] = 0.6n = \frac{3n}{5}.$$

If Alice loses the tournament, then she wins at most half of the games. Thus the losing event is contained in

$$\{X \leq n/2\}.$$

When n is odd, this means strictly fewer than half. When n is even, this event also includes the tie case, so it still gives a valid upper bound.

Now rewrite the threshold $n/2$ in the form $(1 - \delta)\mu$:

$$(1 - \delta)\mu = \frac{n}{2}.$$

Since $\mu = 3n/5$, we get

$$(1 - \delta)\frac{3n}{5} = \frac{n}{2} \implies 1 - \delta = \frac{5}{6} \implies \delta = \frac{1}{6}.$$

Applying the lower-tail Chernoff bound,

$$\Pr[X \leq (1 - \delta)\mu] \leq \exp\left(-\frac{\mu\delta^2}{2}\right).$$

Substitute $\mu = 3n/5$ and $\delta = 1/6$:

$$\Pr[X \leq n/2] \leq \exp\left(-\frac{(3n/5)(1/36)}{2}\right) = \exp\left(-\frac{n}{120}\right).$$

Therefore,

$$\Pr[\text{Alice loses the tournament}] \leq e^{-n/120}.$$

So the probability that Alice loses decreases exponentially fast in n .

Problem 2. In an election with two candidates using paper ballots, each vote is independently misrecorded with probability $p = 0.02$. Use a Chernoff bound to upper-bound the probability that more than 4% of the votes are misrecorded in an election of 1,000,000 ballots.

Solution. Let X be the number of misrecorded votes. Then

$$X \sim \text{Bin}(1,000,000, 0.02).$$

Hence

$$\mu := \mathbb{E}[X] = 1,000,000 \cdot 0.02 = 20,000.$$

The event “more than 4% are misrecorded” means

$$X > 0.04 \cdot 1,000,000 = 40,000.$$

Since $40,000 = 2\mu$, this is the event

$$X > (1 + 1)\mu.$$

Thus we may apply the Chernoff bound

$$\Pr[X \geq (1 + \delta)\mu] \leq e^{-\mu\delta^2/3}, \quad 0 \leq \delta \leq 1.$$

Here $\delta = 1$, so

$$\Pr[X \geq 40,000] \leq e^{-20,000/3}.$$

Since

$$\{X > 40,000\} \subseteq \{X \geq 40,000\},$$

we obtain

$$\boxed{\Pr[\text{more than 4\% misrecorded}] \leq e^{-20,000/3}.$$

Alternatively, one may use the sharper Chernoff form

$$\Pr[X \geq (1 + \delta)\mu] \leq \left(\frac{e^\delta}{(1 + \delta)^{1+\delta}} \right)^\mu.$$

With $\delta = 1$ and $\mu = 20,000$, this gives

$$\Pr[X \geq 40,000] \leq \left(\frac{e}{4} \right)^{20,000}.$$

This is slightly stronger than $e^{-20,000/3}$, but either bound is fully valid.

Problem 3(a). Suppose n balls are thrown independently and uniformly at random into n bins. Find

$$\Pr[\text{bin 1 has exactly one ball} \mid \text{exactly one ball fell into bins 1,2,3}].$$

Solution. Let

$$A = \{\text{bin 1 has exactly one ball}\}, \quad B = \{\text{exactly one ball fell into bins 1,2,3}\}.$$

We want $\Pr[A \mid B]$.

By definition,

$$\Pr[A \mid B] = \frac{\Pr[A \cap B]}{\Pr[B]}.$$

First compute $\Pr[B]$. The event B means:

- exactly one of the n balls goes into the set $\{1, 2, 3\}$, and
- all other $n - 1$ balls go into bins $4, 5, \dots, n$.

For a fixed ball, the probability of landing in one of the first three bins is $3/n$, and the probability of landing outside the first three bins is $(n - 3)/n = 1 - 3/n$. Therefore,

$$\Pr[B] = \binom{n}{1} \frac{3}{n} \left(1 - \frac{3}{n}\right)^{n-1} = n \cdot \frac{3}{n} \left(1 - \frac{3}{n}\right)^{n-1} = 3 \left(1 - \frac{3}{n}\right)^{n-1}.$$

Now compute $\Pr[A \cap B]$. For both A and B to happen, there must be exactly one ball in bin 1 and no balls in bins 2 and 3. Thus:

- choose which one of the n balls lands in bin 1,
- that ball lands in bin 1 with probability $1/n$,
- all other $n - 1$ balls avoid bins 1,2,3 with probability $(1 - 3/n)^{n-1}$.

Hence

$$\Pr[A \cap B] = \binom{n}{1} \frac{1}{n} \left(1 - \frac{3}{n}\right)^{n-1} = \left(1 - \frac{3}{n}\right)^{n-1}.$$

Therefore,

$$\Pr[A \mid B] = \frac{\left(1 - \frac{3}{n}\right)^{n-1}}{3 \left(1 - \frac{3}{n}\right)^{n-1}} = \frac{1}{3}.$$

So the answer is

$$\Pr[\text{bin 1 has one ball} \mid \text{exactly one ball in bins 1,2,3}] = \frac{1}{3}.$$

This also matches intuition: given that exactly one ball landed among bins 1, 2, and 3, symmetry says it is equally likely to be in any of those three bins.

Problem 3(b). Suppose n balls are thrown independently and uniformly at random into n bins. Find the conditional expectation of the number of balls in bin 1 under the condition that bin 2 received no balls.

Solution. Let X_1 be the number of balls in bin 1, and let

$$C = \{\text{bin 2 received no balls}\}.$$

We want to compute

$$\mathbb{E}[X_1 \mid C].$$

Write

$$X_1 = \sum_{j=1}^n I_j,$$

where

$$I_j = \begin{cases} 1, & \text{if ball } j \text{ goes to bin 1,} \\ 0, & \text{otherwise.} \end{cases}$$

Then, by linearity of conditional expectation,

$$\mathbb{E}[X_1 \mid C] = \sum_{j=1}^n \mathbb{E}[I_j \mid C].$$

Since I_j is an indicator,

$$\mathbb{E}[I_j \mid C] = \Pr[I_j = 1 \mid C].$$

So it remains to compute the conditional probability that a particular ball lands in bin 1, given that no ball lands in bin 2.

Using conditional probability,

$$\Pr[I_j = 1 \mid C] = \frac{\Pr[I_j = 1 \text{ and } C]}{\Pr[C]}.$$

First,

$$\Pr[C] = \left(1 - \frac{1}{n}\right)^n,$$

because every ball must avoid bin 2.

Next, for $I_j = 1$ and C to happen:

- ball j must go to bin 1, which has probability $1/n$,
- all the other $n - 1$ balls must avoid bin 2, which has probability $(1 - 1/n)^{n-1}$.

So

$$\Pr[I_j = 1 \text{ and } C] = \frac{1}{n} \left(1 - \frac{1}{n}\right)^{n-1}.$$

Therefore,

$$\Pr[I_j = 1 \mid C] = \frac{\frac{1}{n} \left(1 - \frac{1}{n}\right)^{n-1}}{\left(1 - \frac{1}{n}\right)^n} = \frac{1}{n \left(1 - \frac{1}{n}\right)} = \frac{1}{n-1}.$$

Hence

$$\mathbb{E}[X_1 \mid C] = \sum_{j=1}^n \frac{1}{n-1} = \frac{n}{n-1}.$$

So the answer is

$\mathbb{E}[\text{number of balls in bin 1} \mid \text{bin 2 received no balls}] = \frac{n}{n-1}.$
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A quick interpretation: once we know bin 2 is forbidden, each ball is effectively distributed uniformly among the remaining $n - 1$ bins, so each ball lands in bin 1 with conditional probability $1/(n - 1)$.

Final answers.

1. $\Pr[\text{Alice loses tournament}] \leq e^{-n/120}$.
2. $\Pr[\text{more than 4\% misrecorded}] \leq e^{-20,000/3}$.
- 3(a). $\Pr[\text{bin 1 has one ball} \mid \text{exactly one ball in bins 1,2,3}] = \frac{1}{3}$.
- 3(b). $\mathbb{E}[X_1 \mid \text{bin 2 received no balls}] = \frac{n}{n - 1}$.